Hugo Marquerie 28/04/2026

Lem convergencia uniforme compactos imp convergencia sucesion

Lema 1. *Sea* $\Omega \subset \mathbb{C}$ *un dominio,* $(f_n)_{n\in\mathbb{N}} \subset \mathcal{C}(\Omega)$ *y* $f\colon \Omega \to \mathbb{C}$ *tales que* $f_n \xrightarrow[n\to\infty]{\Omega} f$*. Sea* $(z_n)_{n\in\mathbb{N}} \subset \Omega$ *tal que* $z_n \xrightarrow[n\to\infty]{} a \in \Omega$*, entonces* $f_n(z_n) \xrightarrow[n\to\infty]{} f(a)$*.*

***Demostración*:** Como $z_n \xrightarrow[n\to\infty]{} a$, entonces $\exists N_0 \in \mathbb{N} : \forall n \geqslant N_0 : z_n \in D(a,1)$. Por tanto, $K := \{z_n : n \geqslant N_0\} \cup \{a\}$ es un subconjunto compacto de Ω. Entonces, $f_n \xrightarrow[n\to\infty]{K} f$, es decir, dado $\varepsilon > 0$, $\exists N \in \mathbb{N} : \forall n \geqslant N : \forall z \in K : |f_n(z) - f(z)| < \varepsilon/2$.

En particular, $\forall n \geqslant \max\{N, N_0\} : |f_n(z_n) - f(z_n)| < \varepsilon/2$ y, como por el `lem-convergencia-uniforme-co` $f \in \mathcal{C}(\Omega)$, $|f(z_n) - f(a)| < \varepsilon/2$. Por tanto, para todo $n \geqslant \max\{N, N_0\}$,

$$|f_n(z_n) - f(a)| \leqslant |f_n(z_n) - f(z_n)| + |f(z_n) - f(a)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \qquad \blacksquare$$

Referenciado en

- Lem Convergencia Uniforme Compactos Subsubsucesiones

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